05 The Neighbours Room

Interspecific competition: Lotka–Volterra

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 34 min	**+ 30 min**	The Crowding Room (02): each species here is logistic on its own.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Neighbours tab

Two species, each self-limiting as in the Crowding Room, now also feel each other. One coefficient per species, α, says how much a competitor counts in the currency of one's own crowding. This guide uses the phase plane to predict the outcome before the run, and the invasion criteria to explain it.

LEARNING OBJECTIVES

After the core path you should be able to

- Describe the four outcomes of Lotka–Volterra competition.
- Read a phase plane with two isoclines, and predict the outcome from their arrangement.
- Explain why growth rates change how fast the outcome arrives but not what it is.
- Explain founder control: why whoever arrives first can keep the site.

The extension adds

- Use the invasion criteria to predict outcomes from parameters alone.
- Calculate the coexistence equilibrium and check it against a run.
- Show that coexistence requires $\alpha_{AB}\alpha_{BA} < 1$ when the Ks are equal, and explain what that means biologically.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Problem class:** step 5 on paper, then each group designs a parameter set for a given outcome and tests it.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Stable coexistence

8 min

SET UP

Preset **Stable coexistence**: both $r = 0.5$, both $K = 500$, both $\alpha = 0.60$, both starting at 50.

PREDICT FIRST

Will both species persist? Will each reach its own K, or settle below it?

RUN

Press **Run**.

LOOK AT

the dashed K lines, and the *Phase plane & isoclines*: where the trajectory ends.

EXPLAIN

What is the gap between each curve and its dashed line? Why does a competitor that counts for only 0.6 of one's own kind allow both to persist?

2 Exclusion

10 min

SET UP

Preset **A excludes B**: $K_A = 700$, $K_B = 400$, $\alpha_{AB} = 0.5$, $\alpha_{BA} = 1.8$.

PREDICT FIRST

Look at the isoclines in the phase plane before running. Which species will win?

RUN

Run, then try **B excludes A**.

EXPLAIN

How does the arrangement of the two isoclines tell you the outcome?

3 Founder control

10 min

SET UP

Preset **Founder control**: both $\alpha = 1.6$, both $K = 500$.

PREDICT FIRST

Who will win? Does it depend on the starting numbers?

RUN

Run. Then click at several points in the phase plane to restart from different pairs of abundances.

EXPLAIN

Why can neither species invade the other here? What does this suggest about the order in which species arrive at a site?

4 Speed against outcome

6 min

SET UP

Back to **A excludes B**. Change only r_B: make it 1.5 and r_A 0.1.

PREDICT FIRST

Will the faster-growing B now win?

RUN

Run.

EXPLAIN

What do the growth rates change, and what do they not change?

EXTENSION · MSC

5 Invasion criteria

12 min

SET UP

Preset **A excludes B**.

PREDICT FIRST

Species i can invade a world holding j at its carrying capacity if $1 - \alpha_{ij}K_j/K_i > 0$. Calculate this for A invading B, and for B invading A. Then design a set of K and α values with $\alpha_{AB} = \mathbf{1.2}$ for which both species coexist.

RUN

Read the *Invasion criteria* panel to check your calculation, then enter your design and run it.

EXPLAIN

Coexistence requires $\alpha_{AB} < K_A/K_B$ and $\alpha_{BA} < K_B/K_A$. Why does that force $\alpha_{AB}\alpha_{BA} < 1$?

6 The equilibrium

10 min

SET UP

Preset **Stable coexistence**.

PREDICT FIRST

Solve for $N_A^* = (K_A - \alpha_{AB}K_B)/(1 - \alpha_{AB}\alpha_{BA})$, and the same for B.

RUN

Run to the end and read the populations.

EXPLAIN

What happens to the equilibrium as α approaches 1 from below?

7 On the knife edge

8 min

SET UP

Preset **Founder control**.

PREDICT FIRST

Somewhere in the phase plane is a line separating starts that end with A from starts that end with B. Where would you expect it to run, given the symmetric parameters?

RUN

Click points close to the diagonal $N_A = N_B$ to find the boundary.

EXPLAIN

What is a saddle point, and what does it mean for a real community to sit near one?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Two species of barnacle share a rocky shore. What observations would tell you whether their interaction resembles the “stable coexistence” or the “founder control” preset?
2. BSc Why does $\alpha < 1$ mean the two species occupy different niches?
3. BSc An invasive species grows faster than the native one. Does that mean it will replace it? Use step 4.
4. MSc Lotka–Volterra competition is phenomenological. What does that mean, and which room replaces it with a mechanism?
5. MSc Gause’s experiments with *Paramecium* are often cited as evidence of competitive exclusion. Summarise them and say what this model adds.
6. MSc How might environmental variation through time allow a species that “should” be excluded to persist?

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** Both persist, each below its own K, at about 312 individuals each: the cost of having a neighbour.
- **Step 2.** A wins and settles at $K_A = 700$. When one species' isocline lies entirely outside the other's, that species wins.
- **Step 3.** Whoever starts ahead wins (A, from 60 against 50); clicking elsewhere can flip the result. Each species suppresses the other more than itself, so neither can invade.
- **Step 4.** A still wins, just more slowly: rates change the speed, not the outcome.
- **Step 5.** A invading B: $1 - 0.5 \times 400/700 \approx 0.71 > 0$, yes. B invading A: $1 - 1.8 \times 700/400 = -2.15$, no. With $\alpha_{AB} = 1.2$, coexistence needs $K_A/K_B > 1.2$ and $\alpha_{BA} < K_B/K_A$ — for example $K_A = 600$, $K_B = 400$, $\alpha_{BA} = 0.5$.
- **Step 6.** $(500 - 300)/(1 - 0.36) = 312.5$ for each. As α approaches 1 the equilibrium becomes ever more sensitive and coexistence ever more fragile.
- **Step 7.** By symmetry the boundary runs along the diagonal, with a saddle point at the coexistence equilibrium; a community near a saddle can tip either way from a small disturbance.